Problem 15.4

Calculate the present value of 300 paid at the end of each year for 20 years using an annual effective interest rate of 8%.

Solution

The payments are made at the end of each year, so this is an annuity-immediate. The annual effective interest rate is

$$i = 0.08,$$

and the number of payments is

$$n = 20.$$

The present value is

$$PV = 300a_{\overline{20}|}.$$

Using the annuity-immediate formula,

$$a_{\overline{20}|} = \frac{1 - v^{20}}{i},$$

where

$$v = \frac{1}{1+i}.$$

Thus,

$$PV = 300\left(\frac{1 - (1.08)^{-20}}{0.08}\right).$$

Finally,

$$PV \approx 2945.4422.$$

Hence, the present value is

$$\boxed{2945.44}.$$